Precision as a software choice: FP64, double-double, triple-double and beyond from the INT8 tensor cores of a laptop GPU

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Abstract

Consumer and laptop GPUs execute FP64 at 1/64 of their FP32 rate, while their INT8 tensor cores are several hundred times faster than their FP64 units. On a laptop Blackwell GPU (RTX PRO 3000, sm_120) we measure 232 TOPS of INT8 `mma.sync` against 0.449 TFLOPS of native FP64, a ratio of 517×. Building on the Ozaki scheme II—exact integer products of residues and reconstruction by the Chinese remainder theorem—we implement, in CUDA C and inline PTX, matrix multiplication at FP64 (5.56 TFLOPS, 12.4× the native peak and more accurate), double-double (∼106 bits, 1.76 TFLOPS), triple-double (∼159 bits, 1.58 TFLOPS) and arbitrary precision (212, 318 and 424 bits) from the same INT8 units. To our knowledge no GEMM beyond FP64 had been built this way. The higher precisions pay off in a real task: solving $Ax = b$ with $\kappa(A) = 10^{20}$ in double-double takes 0.855 s at $n = 8192$, 499× sooner than a `__float128` LU on all 24 CPU threads of the same laptop and with an error as small; at $\kappa = 10^{36}$, where FP64 and double-double have no correct digit, a triple-double LU on the GPU is 32× faster than the same arithmetic on 24 CPU threads and 145× faster than MPFR, and is the most accurate of the three. Counting CPU and GPU power, a double-double solve uses ∼270× less energy than the quad-precision CPU solve. Leaving 8-bit moduli for primes below 2^{16} and 2^{23} split into bytes, an *exact* product of integer matrices with entries of 1 015 to 131 063 bits costs work linear in the bits, is bit-exact against GMP, and runs 10–19× faster than FLINT's multimodular `fmpz_mat_mul` on 24 CPU threads, with 15–18× less energy. Turning the problem around—int8 in, residues out—the exact determinant of an int8 matrix comes 224× sooner than FLINT at $n = 4096$ and 355× at $n = 8192$ (98 139 bits), with a proved bound on $|\det A|$ tight to one bit, and $n = 16\,384$ (204 480 bits) takes 66 s, $n = 32\,768$ (425 348 bits) 594 s within the 12 GB of the GPU. For entries of 32 to 256 bits, high-order lifting on the same modular GEMMs (the residue squared in RNS, the solution vector carried through INT8 GEMMs) and a proved bound for byte-plane entries give the determinant 236× sooner than FLINT at $n = 2048$ with 64-bit entries, 306× with 128-bit entries and 265× at $n = 4096$ (276 398 bits in 3.3 s against 868 s): the ratio grows with the entry size. The laptop is power-bound (∼95 W) under these kernels, which shapes what can and cannot be gained. On 13 of the most ill-conditioned real matrices of NIST Matrix Market ($n \leq 5005$, entries up to 154 bits) the certified exact solution takes 0.2–34 s, up to ∼900× sooner than SPEX; FP64 has no correct digit on 7 of them. The same units give correctly rounded MXFP4 and BF16 products: on the real weights of Kimi K3 (2.78 trillion parameters, the largest open-weights model) every output of 48 expert matrices is certified and identical for any batch, where BF16 tensor cores with FP32 accumulation round 51–83% of them differently; at decode the exact kernel reads 94 GB/s, 18× the laptop's NVMe, so reproducibility costs nothing when such a model streams from disk.

Përmbledhje (shqip). Kartelat grafike të laptopëve e ekzekutojnë FP64 me 1/64 e shpejtësisë FP32, ndërsa tensor core-t INT8 janë qindra herë më të shpejtë. Mbi një GPU Blackwell laptopi matëm 232 TOPS INT8 kundrejt 0,449 TFLOPS FP64 (517×). Me skemën Ozaki II (prodhime të sakta të mbetjeve dhe teorema kineze e mbetjeve) ndërtuam shumëzim matricash me saktësi FP64, double-double, triple-double dhe arbitrare (deri 424 bit) nga të njëjtat njësi INT8. Te një punë reale—$Ax = b$ me $\kappa = 10^{20}$—zgjidhja double-double mbaron 499× më shpejt se LU

Table 1: Peak throughput of the laptop GPU (`bench/peak.cu`).

Unit	Throughput
FP64 FMA, native	0.449 TFLOPS
FP32 FMA, native	28.2 TFLOPS
INT8 `mma.sync.m16n8k32` (PTX)	232 TOPS
INT8 / FP64	**517×**

`__float128` mbi 24 fije CPU, me gabim po aq të vogël, dhe me ∼270× më pak energji. Me prime nën 2^{16} e 2^{23} të ndara në bajte, shumëzimi i *saktë* i matricave me numra të plotë 1 015–131 063 bitësh del bit për bit i saktë, 10–19× më shpejt se FLINT dhe me 15–18× më pak energji. Përcaktori i saktë i një matrice int8 del 224× më shpejt se FLINT te $n = 4096$ dhe 355× te $n = 8192$, me një kufi të provuar të ngushtë brenda një biti; te $n = 32\,768$ (425 348 bit) mbaron për 594 s brenda 12 GB të GPU-së. Për hyrje 32–256 bit, ngritja e rendit të lartë mbi të njëjtat GEMM modulare dhe një kufi i provuar për hyrje me plane bajtesh e japin përcaktorin 236× më shpejt se FLINT te $n = 2048$ me hyrje 64-bit, 306× me 128-bit dhe 265× te $n = 4096$; raporti rritet me madhësinë e hyrjeve.

1 Introduction

The FP64 units of consumer and laptop GPUs are a small fraction of the chip: on our laptop Blackwell part a peak kernel reaches 0.449 TFLOPS of FP64 against 28.2 TFLOPS of FP32 and 232 TOPS of INT8 tensor-core throughput (Table 1). Ozaki, Uchino and Imamura showed that FP64 matrix multiplication can be emulated on INT8 tensor cores (Ozaki scheme II): each input is scaled to integers, reduced modulo a set of pairwise coprime moduli, the residue matrices are multiplied exactly in INT8 with INT32 accumulation, and the integer product is reconstructed by the Chinese remainder theorem (CRT) before being scaled back [5]. FP64 by this route is not new; it is our baseline.

What we add is the observation, made concrete and measured, that the same machinery does not stop at FP64. The precision is set by the number of moduli and by how many floating-point words each input and output carries; the expensive part, the INT8 GEMMs, runs at the same speed whatever the precision. This turns precision into a software choice on hardware that has no wide floating-point unit at all. We build and measure:

- FP64 GEMM at 16 moduli: 5.56 TFLOPS, 12.4× the native FP64 peak, with smaller error than native FP64;
- double-double (∼106-bit) GEMM at 32 moduli: 1.76 TFLOPS, 3.9× the native FP64 peak with twice the bits;
- triple-double (∼159-bit) GEMM at 49 moduli—after which every prime below 256 divides one of them, so no further 8-bit modulus can join—at 1.58 TFLOPS;
- arbitrary precision beyond that from balanced 160-bit slices: 212, 318 and 424 bits;
- dense linear solves at condition numbers where FP64 and double-double fail, all on the GPU.
- exact determinants of int8 matrices up to $n = 32\,768$, and of matrices with 32–256-bit entries by high-order lifting on tensor cores, 63–306× faster than FLINT.

Everything runs on one laptop, and all competitors were measured on the same machine.

2 Method

Ozaki scheme II. For $C = AB$ the rows of A and columns of B are scaled by powers of two to integers A', B' whose product cannot exceed $M/2$ in magnitude, where $M = \prod_\ell m_\ell$ is the product of L pairwise coprime moduli $m_\ell \le 256$. For each ℓ the residues $A' \bmod m_\ell$ and $B' \bmod m_\ell$ fit in INT8, their product is computed exactly by the tensor cores with INT32 accumulation, and reduced mod m_ℓ. The CRT then gives $A'B'$ exactly as a signed integer of up to $\log_2 M$ bits, which is scaled back. The precision of the result is limited only by how many bits of each input survive the scaling, i.e. by $\log_2 M$.

Wider inputs and outputs. For double-double and triple-double inputs, each of the two or three parts is split exactly into 53-bit integer pieces before the residues are formed; like pieces are summed and cut once into 16-bit chunks, so the residue of a wide integer costs 32-bit multiply-adds per modulus instead of 64-bit reductions (triple-double residues $57 \to 19$ ms at $n = 4096$). On the output side the integer is never converted through floating-point arithmetic: the CRT is evaluated as a direct sum $x = \sum_\ell r_\ell W_\ell - qM$ over 32-bit limbs on the integer units, with q from 40-bit fixed-point fractions, and the result is peeled off the integer as two or three doubles (round down, round down, round to nearest, each exact from the integer), then normalized. This replaced a Garner reconstruction followed by a double-double Horner evaluation, which took 41.5 of 47 ms of a 4096^2 double-double GEMM because FP64 runs at 1/64 rate on this chip; the direct sum costs 0.68 ns per output element instead of 2.8 and is nearly independent of K. It was checked on the host against 10^5 random 200-bit integers with no error.

Moduli. FP64 uses 16 moduli, double-double 32, triple-double 49 (the 32 plus the 17 primes $103 \ldots 29$ that are not yet factors; $M \approx 2^{342}$). With these, every prime below 256 divides one of the moduli, so no further 8-bit modulus coprime to all of them exists and M cannot grow within one GEMM. Beyond ∼160-bit inputs we cut each input into balanced 160-bit slices and run one 49-modulus GEMM per slice pair that reaches the result, accumulating exactly in wide integers and peeling Q doubles.

3 Accuracy

Errors are the maximum componentwise $|C - C_{\text{ref}}|/(|A||B|)$ in units of the target precision's unit roundoff, against higher-precision CPU references (80-bit for FP64, `__float128` for double-double, an exact 600-bit MPFR sum for triple-double, an exact 2400-bit sum beyond). Inputs have magnitudes spread over $2^{\pm 0}$ or $2^{\pm 8}$.

Table 2: Error in units of 2^{-53} (FP64, $M = N = 256$) and 2^{-106} (double-double, $M = N = 128$).

K	magnitudes	native FP64	14 moduli	**16 moduli**
512	$2^{\pm 0}$	4.71	10.9	**0.99**
512	$2^{\pm 8}$	24.1	395	**3.1**
4096	$2^{\pm 0}$	4.15	10.5	**0.25**
4096	$2^{\pm 8}$	25.1	203	**1.87**
K	magnitudes	24 moduli	28 moduli	**32 moduli**
256	$2^{\pm 0}$	$6.2 \cdot 10^5$	27.1	**1.09**
256	$2^{\pm 8}$	$2.0 \cdot 10^7$	1160	**46.9**
1024	$2^{\pm 0}$	$4.7 \cdot 10^5$	31	**0.49**
1024	$2^{\pm 8}$	$1.1 \cdot 10^7$	713	**12.4**

Table 3: Triple-double error in units of 2^{-159}, $M = N = 128$, against an exact MPFR sum.

K	magnitudes	41 moduli	45 moduli	**49 moduli**
256	$2^{\pm 0}$	$1.7 \cdot 10^4$	5.2	**0.24**
256	$2^{\pm 8}$	$1.2 \cdot 10^6$	343	**10.7**
1024	$2^{\pm 0}$	$1.8 \cdot 10^4$	4.6	**0.13**
1024	$2^{\pm 8}$	$4.3 \cdot 10^5$	110	**2.1**

Beyond triple-double, against an exact 2400-bit sum at $K = 256$: 0.13 ulp at 212 bits, 0.25 at 318 bits, 0.08 at 424 bits.

4 Speed

Table 4: GEMM speed on the laptop. CPU numbers with the CPU otherwise idle, measured at $n = 256/512$ and scaled by n^3.

bits	n	GPU time	GPU TFLOPS	best CPU code, 24 threads	MPFR, 24 threads
53 (FP64)	4096	—	5.56 (12.4× native)	—	—
106 (dd)	4096	—	1.76	—	—
159 (td)	4096	87 ms	1.58	td AVX ∼10.7 s (123×)	∼208 s (2390×)
212	2048	36 ms	0.47	QD `qd_real` ∼16.2 s (445×)	∼35 s (960×)
318	2048	52 ms	0.33	—	∼37 s (713×)
424	2048	80 ms	0.22	—	∼41 s (514×)

At 16 moduli the time of a 4096^3 FP64 GEMM splits into scales 1.0 ms, residues 3.0 ms, the 16 INT8 GEMMs 16.8 ms and the CRT 3.6 ms. The INT8 GEMM reaches 157 TOPS, 67% of the measured peak, with a three-stage `cp.async` pipeline; a faster kernel did not move end-to-end times, because with real data the laptop is power-bound (Section 10).

5 A real task: ill-conditioned linear systems

We solve $Ax = b$ with $A = Q_1 \operatorname{diag}(\sigma) Q_2$, $Q_{1,2}$ orthogonal (a DST-I, the second permuted and sign-flipped), σ spaced from 1 to 10^{-20} or 10^{-36}; such matrices cannot be fixed by scaling. $b = Ax_{\text{true}}$ is summed exactly. The error is $\max |x - x_{\text{ref}}| / \max |x_{\text{ref}}|$.

Double-double, $\kappa = 10^{20}$. κu is 10^4 for FP64, so FP64 LU has no correct digit and textbook iterative refinement diverges (it needs $\kappa u < 1$; at $\kappa = 10^{12}$ the same refinement converges to $3 \cdot 10^{-20}$, so the baseline is right). The GPU solver keeps everything on the device: a recursive panel with cooperative 32-column leaves (a thread per row holding the row in registers, rows renamed instead of swapped, one grid synchronization per column), halves joined by double-double GEMMs on the tensor cores, L_{11}^{-1} assembled from the halves' inverses, left-looking solves with inverted diagonal blocks, and one refinement step in double-double.

The 0.956 s at $n = 8192$ was measured in the same run as the quad reference, after minutes of all-core load; alone on a cool GPU the solve takes 0.855 s. Of those, the panel takes 0.45 s (its leaves 0.22 s, bound by the FP64 units at ∼10 GFLOPS per SM), the trailing GEMMs 0.15 s, solves and refinement 0.09 s, U_{12} 0.04 s.

Triple-double, $\kappa = 10^{36}$. Here κu is 10^{20} for FP64 and 10^4 for double-double: neither has a correct digit. The same solver structure in three parts per number, with 16-column leaves:

Table 5: Double-double LU on the GPU vs `__float128` LU on 24 CPU threads, $\kappa = 10^{20}$, nb = 1024.

n	dd LU, GPU	`__float128` LU	speedup	error vs x_{true} (ours / quad)
2048	0.113 s	7.65 s	67×	$1.6 \cdot 10^{-11}$ / $2.4 \cdot 10^{-11}$
4096	0.283 s	55.6 s	$\sim$ 197×	—
6144	0.572 s	200.8 s	351×	$3.4 \cdot 10^{-10}$ / $2.9 \cdot 10^{-10}$
8192	0.956 s	476.9 s	**499**×	$2.1 \cdot 10^{-11}$ / $2.3 \cdot 10^{-11}$

Table 6: Triple-double LU, $\kappa = 10^{36}$; GPU speedups in parentheses; CPU otherwise idle.

n	**td LU, GPU**	td LU, 24 CPU threads	MPFR LU, 160 bits	FP64 LU
1024	**0.082 s**, $1.3 \cdot 10^{-14}$	0.10 s, $3.6 \cdot 10^{-14}$ (1.2×)	1.32 s, $3.3 \cdot 10^{-14}$ (16×)	128
2048	**0.212 s**, $2.1 \cdot 10^{-14}$	1.04 s, $7.6 \cdot 10^{-14}$ (5×)	10.2 s, $5.6 \cdot 10^{-14}$ (48×)	103
4096	**0.563 s**, $2.2 \cdot 10^{-14}$	18.0 s, $1.2 \cdot 10^{-13}$ (32×)	81.9 s, $9.6 \cdot 10^{-14}$ (145×)	86

At $n \le 1024$ a well-vectorized triple-double LU on 24 cores keeps up; the gap opens with n as the work moves into the tensor-core GEMMs. At $n = 4096$ the panels take 0.38 s of the 0.563 s (leaves 0.14 s): the panel, not the GEMM, is now the limit.

6 Thousands of bits, exactly: linear in the bits

The 49 moduli ≤ 256 cap one Ozaki-II product at $\sim$160-bit inputs, and slicing beyond that costs a number of GEMMs that grows with the square of the slices. For wide integers we leave 8-bit moduli: the moduli are primes $p < 65\,280$, and each balanced residue ($|r| < p/2$) is split into two signed bytes, $r = 256h + \ell$. Then $AB \bmod p$ is four exact INT8 GEMMs (hh, $h\ell$, ℓh, $\ell\ell$) recombined in 64-bit integers, so *the GEMM work grows linearly with the number of bits.* Above $\sim$60 000 bits of the CRT modulus the residues take three bytes and the primes run up to 2^{23} (nine products per prime, $\sim$23 bits each). The two conversions are INT8 GEMMs as well, as conversions to and from a residue number system are matrix products: the residues are the balanced byte digits of each entry times the bytes of $2^{8j} \bmod p$; for the CRT, $t_\ell = r_\ell\,(M/p_\ell)^{-1} \bmod p_\ell$ (bytes) times the bytes of M/p_ℓ gives the digits of $x + qM$, with $q = \text{round}(\sum_\ell t_\ell/p_\ell)$ from doubles, and qM is taken off while the digits are carried. Everything runs on the tensor cores (`src/ozaki_big.cuh`, `bench/big_test.cu`).

Every product is exact: 256 sampled elements per run agree bit for bit with GMP. The baselines compute the same exact product on all 24 CPU threads: GMP inner products (`mpz_addmul`), and FLINT's `fmpz_mat_mul`, which is itself multimodular with BLAS-style kernels and is the strongest baseline we know of.

entry bits	n	primes (bytes)	GPU	GMP	FLINT	GPU vs FLINT
1 015	1024	192 (2)	71 ms	23.0 s	1.31 s	18×
4 087	512	576 (2)	62 ms	16.4 s	1.07 s	17×
16 375	256	2 112 (2)	90 ms	14.0 s	1.69 s	19×
4 087	1024	640 (2)	341 ms	125 s	6.10 s	18×
16 375	512	2 112 (2)	363 ms	104 s	6.33 s	17×
32 759	256	4 352 (2)	316 ms	36.1 s	4.56 s	14×
65 527	128	5 760 (3)	313 ms	10.6 s	3.45 s	11×
131 063	128	11 520 (3)	983 ms	29.6 s	9.76 s	10×

GMP times are measured on a corner of the outputs (96^2, or 48^2 above 8 000 bits) and scaled; FLINT runs the full product. The largest entries have $\sim$39 450 decimal digits. Energy per

exact product (RAPL package + `nvidia-smi`, both chips counted in both columns): 1 015 bits, $n = 1024$: 8.6 J on the GPU path against 131 J for FLINT (15×); 16 375 bits, $n = 256$: 7.9 J against 142 J (18×).

Doubling n at fixed width does not raise the ratio either (16 375 bits: 19× at $n = 256$, 17× at $n = 512$): even at $n = 512$ the conversion GEMMs ($\sim 1.9 \cdot 10^{13}$ INT8 operations) dwarf the 8 704 product GEMMs ($\sim 2.3 \cdot 10^{12}$). The ratio to FLINT falls slowly beyond ∼16 000 bits (19× → 10×) because the conversions are n^2P^2 work for P-bit entries while the products are n^3P: at $n = 128$ the conversions dominate. The way past that crossover is a subquadratic conversion (a two-level residue system), which we leave for the next version. For scale: one part in a googol (10^{-100}) is ∼333 bits, already inside the 424-bit format of the previous section.

7 Changing the problem: exact determinants and exact solutions

The ratio of the previous section falls with the bits because large numbers must enter and leave the residue system. Many exact computations produce huge numbers from small inputs: the determinant of an $n \times n$ int8 matrix has ∼0.72 bits per row less than Hadamard's bound, about 47 000 bits at $n = 4096$, and the rational solution of $Ax = b$ has n numerators of that size. There the conversions vanish: the input is int8, each prime returns one residue (or a vector of digits), and the only big-number step is on the host (`src/ozaki_det.cuh`, `bench/det.cu`).

LU modulo thousands of primes. For primes $p < 65\,280$ and residues X, W, with $X = X_\ell + 256X_h$ in balanced bytes and $V = 256W \bmod p$, $XW \equiv [X_\ell\, X_h][W_\ell; V_\ell] + 256\,[X_\ell\, X_h][W_h; V_h] \pmod p$: two INT8 GEMMs of depth $2K$, reduced once in the epilogue. A recursive LU (Toledo) without pivoting runs for a batch of primes at once, every $O(n^3)$ step such a product. Its 64-column leaves invert their 64×64 diagonal blocks in shared memory. A prime with a vanishing leading minor (probability $\sim n/p$) is dropped. Structured matrices, whose leading minors vanish over $\mathbb{Z}$, are preconditioned as UAL with random unit triangular U, L (Kaltofen–Saunders) whose entries in $[0, 127]$ are the same residues for every prime.

A divisor by p-adic lifting. (The divisor-then-primes hybrid is classical on CPUs [17]; FLINT uses it.) $A^{-1} \bmod p$ comes from the LU of $[A\ I]$. Dixon lifting runs on the GPU, one 16-bit and one int8 matrix–vector product per step: $x_i = A^{-1} r_i \bmod p$, $r_{i+1} = (r_i - Ax_i)/p$. Rational reconstruction of a few entries of $x = A^{-1}b$ gives d, which divides $\det A$ and for random b is almost always the largest invariant factor. Then $\det A/d$ needs only enough primes to exceed a bound on $|\det A|/d$.

A proved bound tight to one bit. Hadamard's bound leaves $\sim 0.72n$ bits of slack on a random matrix (5 900 at $n = 8192$), i.e. hundreds of primes for $\det A/d$. We replace it by a certificate in the spirit of Ogita's verified determinants [16]. FP32 LU with partial pivoting gives $PA \approx LU$. $X_L = \lfloor 2^s L^{-1} \rceil$ (unit diagonal, so exactly 2^s) and $X_U = \lfloor 2^s U^{-1} \operatorname{diag}(u) \rceil$ are integer triangular matrices, so their determinants are products of their diagonals. $B = X_L\, PA\, X_U$ is computed *exactly* (the same modular INT8 GEMMs, int64 operands reduced while packing, Garner to 128 bits), and Hadamard's inequality on B, accumulated with upward rounding, bounds $|\det A| = |\det B|/(2^{sn}\,|\det X_U|)$. Floating point only decides how close B is to $2^{2s}\operatorname{diag}(u)$, that is how tight the bound is, never whether it holds. Measured: 47 021 bits against $|\det A|$ of 47 021 bits at $n = 4096$, 98 140 against 98 139 at $n = 8192$. $\det A/d$ then needs 6 primes instead of 224 and 496.

Results. Random int8 matrices, FLINT 3.0.1 on all 24 CPU threads; every result bit-exact against FLINT. FLINT's `fmpz_mat_det` is already its accelerated path (a Dixon divisor, then

primes). Its plain multimodular `fmpz_mat_det_modular` takes 347 s at $n = 2048$, 177× our all-primes LU with the same algorithm.

n	$\lvert\det A\rvert$ bits	GPU, all primes	GPU, lifting + bound	FLINT	vs FLINT
1 024	10 736	0.25 s	0.073 s	2.89 s	40×
2 048	22 485	1.96 s	0.293 s	24.9 s	85×
4 096	47 021	21.3 s	1.15 s	257 s	224×
8 192	98 139	—	8.05 s	2 861 s	355×
16 384	204 480	—	65.9 s	—	
32 768	425 348	—	594 s	—	

At $n = 8192$ FLINT ran while some GPU runs shared the CPU; the leading digits agree, and $n = 8192, 16\,384, 32\,768$ are also checked modulo four primes used nowhere else. The exact rational solution of $Ax = b$ (same lifting; all n entries back to integers by Horner in halves on 24 threads; one common denominator) is bit-exact against FLINT's `fmpz_mat_solve_dixon_den`: 0.064 s against 2.33 s at $n = 1024$ (36×), 0.145 s against 18.9 s at $n = 2048$ (130×), 1.62 s against 145 s at $n = 4096$ (90×, 192.6 Mbit of numerators), and 7.5 s at $n = 8192$ (804 Mbit). Energy per determinant at $n = 2048$, both chips counted: ∼40 J on the GPU path (process start-up included; ∼23 J for the computation alone) against ∼2 030 J for FLINT, ∼50× less. At $n = 8192$ the lifting is the cost: 13 013 steps, each reading A^{-1} (16 bits per entry) and A, at ∼100% of the read bandwidth we measure on this GPU (∼410 GB/s). That limit is physical for digit-by-digit lifting; moving past it needs lifting built on matrix products (high-order lifting, the DoublePlusOneLift of Pauderis and Storjohann [19]), the next step.

$n = 32\,768$ in 12 GB. One modular product packs $(2n + 128) \cdot 2n$ bytes three times (13 GB), and the bound holds three float and two int64 $n \times n$ matrices (30 GB). Products deeper than 2048 now run in K-chunks (a third epilogue mode, $R \mathrel{+}= XW$); the bound builds X_L, X_U in column panels (triangular solves against identity panels) kept in host memory, forms $T = X_L\, PA \bmod q$ in row panels and $B = TX_U$ in column panels as 16-bit residues on the host, and sums the rows in panels, the L_1 norms exactly in 128 bits; it runs on its own stream beside the lifting. Peak use: ∼11 GB of VRAM and 29 GB of host memory. The determinant (425 348 bits) takes 594 s: A^{-1} 15 s, 56 145 lifting steps 497 s at the bandwidth limit, $\det A/d$ 39 primes 71 s; the bound, 244 bits above $\lvert\det A\rvert$ at this size (FP32 LU; a larger scale 2^s does not change it), is hidden under the lifting. Each piece was checked bit-exact against FLINT at $n \le 4096$ with the chunks and panels forced small.

Large entries. With entries of L bytes stored as byte planes ($A = \sum_j 256^j A_j$), the residues come by Horner and the LU is unchanged; the lifting reads the L planes per step and keeps the residual as $L + 4$ base-256 digits per row, divided by p exactly by one lane. Lifting modulo $M = p_0 \cdots p_{S-1}$ ($S \approx L/2$ primes at once: x_i by CRT from S residues, Ax_i on byte-aligned digits, division by each p_t in turn) gains $16S$ bits per read of the L planes and cuts the lifting 1.3–2.9×. Bit-exact against FLINT: $n = 1024$ with 64-bit entries in 0.247 s against 12.4 s (50×), with 128-bit entries 0.692 s against 31.0 s (45×); $n = 2048$ with 32-bit entries 0.563 s against 46.7 s (83×), with 64-bit entries (a 137 173-bit determinant) 2.03 s against 106.9 s (53×). GPU times come from runs without FLINT beside them: CPU and GPU share one power budget, and right after 24 FLINT threads the GPU is up to 2× slower. Each step still reads all L planes, so the cost grows faster than the bits.

High-order lifting. Digit-by-digit lifting reads A and A^{-1} once per X-digit; high-order lifting (Storjohann's sparse inverse expansion [18], in row form $yA = b$) squares the residue instead, so the precision doubles per level. With $X = p_0 \cdots p_{s-1} \ge 6.25\, n^2 \|A\|$ (distinct 16-bit primes, $B_0 = A^{-1} \bmod X$ from the same batched LU), $R_0 = (I - AB_0)/X$ and, per level, $\bar{R} = R_i^2$,

Table 7: Exact determinants of $n \times n$ matrices with random b-bit entries: total GPU time with S-prime Dixon lifting and with high-order lifting (both with the proved bound for byte planes and FLINT's reconstruction), against FLINT `fmpz_mat_det` on 24 threads. GPU runs without FLINT beside them; all bit-exact.

n	b	$\lvert\det A\rvert$ bits	Dixon	HOL	FLINT	vs FLINT
256	256	65 914	0.139 s	0.063 s	1.08 s	17×
512	256	132 082	0.399 s	0.139 s	8.82 s	63×
1024	64	68 073	0.177 s	0.126 s	12.39 s	98×
1024	128	133 615	0.434 s	0.219 s	30.98 s	141×
2048	32	71 638	0.358 s	0.288 s	46.70 s	162×
2048	64	137 173	1.091 s	0.453 s	106.89 s	236×
2048	128	268 249	2.946 s	0.856 s	261.65 s	306×
4096	32	145 330	3.425 s	1.934 s	416.46 s	215×
4096	64	276 398	7.443 s	3.273 s	867.68 s	265×

$M_i = B_0\bar{R} \bmod X$, $R_{i+1} = (\bar{R} - AM_i)/X$ exactly, with $\|R_i\| \le 1.25\,n\|A\|$; then $c_{i+1} = c_i + X^{m_i}c_iR_i + X^{2m_i}bM_i$, $m_{i+1} = 2m_i + 1$, gives $c_i \equiv bA^{-1} \pmod{X^{m_i}}$. The residue chain runs in RNS, modulo the s primes of X and w more primes ($\prod q > 2.5\,n\|A\|$): M_i by Garner from its residues, $R_{i+1} \bmod q = (\bar{R} - AM_i)X^{-1}$, its exact value by Garner over the q (checked against its bound) — $2(s + w)$ modular GEMMs per level and no big number anywhere. All matrices are kept transposed so that the vector chain reads K-contiguous rows. That chain is itself a GEMM: c_i is stored as balanced digits in a periodic mixed radix (position e has radix $p_{e \bmod s}$), so X^m is a shift by sm positions and the Garner digits of M_i are already digits of it; rows are the digits as [lo | hi] bytes, columns the byte planes of R_i as $[Z_v \mid Z_{v-1}]$, so column v collects 256^v; the planes are combined in parallel and the carries run down the positions one thread per entry. Around it, the proved bound now takes byte planes (a float LU of $A/256^{L-1}$ only decides tightness; $B = X_L\,PA\,X_U$ is exact from the planes, Garner over up to 16 primes with $|B_{ij}|$ rounded up from the mixed-radix digits): at $n = 2048$ with 64-bit entries it is 1 bit above $\lvert\det A\rvert$, and $\det A/d$ needs 6 primes instead of 128. The rational reconstruction is FLINT's half-gcd `_fmpq_reconstruct_fmpz_2` (0.49 s → 0.047 s there). Bit-exact against FLINT in every case of Table 7: the lifting is 2.6–4.3× faster than S-prime Dixon and grows like the bits, not their square, so the ratio to FLINT now rises with the entry size—63× at $n = 512$ with 256-bit entries, 236× at $n = 2048$ with 64-bit and 306× with 128-bit entries, 265× at $n = 4096$ with 64-bit entries (a 276 398-bit determinant in 3.27 s against 868 s). At that size the residue chain is 2.08 s of the 2.71 s lifting (247 modular GEMMs of n^3, power-bound), the vector chain 0.44 s. Moduli below 256 for X (one GEMM per modulus, as in Ozaki-II) are not an option here: without pivoting the LU modulo p fails with probability $\approx 1 - e^{-n/p}$.

8 Real systems: how wrong FP64 is, and the exact answer

The residue chain of the lifting above now runs over 8-bit moduli wherever B_0 is not involved: Z^2 exactly in an RNS of 23–42 odd moduli below 256 (one INT8 GEMM each; on random data a 16-bit modular product costs 2.8× more per bit of modulus at $n = 4096$), a base extension by approximate CRT to the 16-bit primes of X, and AM modulo the first 10–19 moduli only. At 4096×64 bits this takes the determinant from 3.22 to 2.28 s (380× FLINT), with the same result.

We then took the 13 most ill-conditioned square real matrices with $900 \le n \le 8192$ of NIST Matrix Market—fluid dynamics (FIDAP, LNS), magnetohydrodynamics (MHD), the Quebec Hydro network, crystal growth, oil reservoirs, a chemical plant—and solved $Ax = b$ *exactly*. Every double is $m\,2^e$, so scaling row i by 2^{k_i} gives integer entries (up to 154 bits) and the same x, with no rounding anywhere; rows are ordered by FP64 partial pivoting, since the GPU LU does not

pivot. High-order lifting gives all n entries, a common denominator follows from a few half-gcd reconstructions, the reconstruction is unique by the Hadamard and Cramer bounds, and $yM = c$ is checked modulo two further primes. On the same integers we ran SPEX Left LU [4] (SuiteSparse 7.6, exact sparse REF LU, one CPU thread); both programs print a digest $\sum_j x_j r^j \bmod 2^{61}-1$. FP64 is LAPACK `dgesv` and SuperLU on the original doubles.

matrix	n	κ (NIST)	bits	GPU s	SPEX s		same x	FP64 digits (entries with none)	
								LAPACK	SuperLU
west0989	989	$5.7{\cdot}10^{12}$	75	0.17	1.5	9×	yes	11.7 (0)	11.8 (0)
qh1484	1484	$5.6{\cdot}10^{17}$	96	0.55	12.4	23×	yes	11.2 (0)	12.2 (0)
fidap032	1159	$5.2{\cdot}10^{20}$	60	0.28	54.5	193×	yes	0.0 (568)	0.0 (572)
fidap006	1651	$3.5{\cdot}10^{21}$	154	1.08	963.7	895×	yes	0.0 (887)	0.0 (693)
cry2500	2500	$4.4{\cdot}10^{17}$	70	1.41	699.3	497×	yes	6.4 (0)	5.1 (0)
fidapm07	2065	$6.8{\cdot}10^{19}$	118	1.19	>7200		–	0.0 (430)	0.0 (290)
fidap026	2163	$4.7{\cdot}10^{18}$	153	2.12	>7200		–	0.0 (1568)	0.0 (1530)
mhd3200a	3200	$6.5{\cdot}10^{39}$	130	4.02	1290.8	321×	yes	0.0 (2876)	0.0 (2768)
fidap014	3251	$2.6{\cdot}10^{16}$	124	4.74	>7200		–	7.6 (0)	8.0 (0)
lns_3937	3937	$1.0{\cdot}10^{17}$	79	4.31	–		–	10.1 (2)	7.4 (1)
fidap012	3973	$2.2{\cdot}10^{20}$	119	7.44	–		–	0.0 (483)	0.0 (482)
mhd4800a	4800	$3.2{\cdot}10^{47}$	131	34.39	–		–	0.0 (3923)	0.0 (4754)
sherman3	5005	$6.9{\cdot}10^{16}$	102	14.25	–		–	11.8 (0)	12.4 (0)
fidapm15	9287	$1.36{\cdot}10^{20}$	63	78.66	–		–	0.0 (2251)	0.0 (5260)
cry10000	10000	$6.9{\cdot}10^{20}$	76	115.02	–		–	4.2 (66)	1.6 (923)

Table 8: Exact solutions of real systems on the laptop (GPU: A^{-1}, lifting, reconstruction and all numerators). SPEX ran afterwards on the CPU alone, limit 2 h. $yM = c$ held modulo both check primes in every row.

On 7 of the 13 systems FP64 has not one correct digit, LAPACK and SuperLU alike—on mhd3200a and mhd4800a the normwise error is 10^8 and 82–99 % of the entries have no correct digit—and NIST's own (FP64) condition estimate does not say which: qh1484 at $5.6 \cdot 10^{17}$ keeps 11 digits. The certified exact solution takes 0.2–34 s; SPEX returns the same x, bit for bit, where it finishes within the limit, 9 to ∼900× later.

9 The largest open model: correctly rounded MXFP4 and BF16 products

Kimi K3 (Moonshot AI; read from its Hugging Face model card and safetensors headers: 2 779 931 837 184 parameters, 1.56 TB) is the largest open-weights model. Its routed experts (2.72 T parameters; 93 layers × 896 experts, 16 per token) are stored in MXFP4: an E2M1 value per weight, one E8M0 scale per 32. A group of 32 is the depth of one `mma.m16n8k32` INT8 step and 2·E2M1 is an integer ≤ 12, so MXFP4×BF16 products can be computed *exactly* on INT8 tensor cores and rounded once (`llm/mxexact.cu`, `llm/exact.cuh`). Weights: $q' = 2\,\mathrm{E2M1}{\cdot}2^{(s-s_{\min}) \bmod 4}$ is an int8 (≤ 96), one scale per row and band of four binades (every row of the 48 K3 matrices tested spans ≤ 3 binades: one band). Activations: each token is aligned to its largest exponent and cut into S int8 digits ($D_c = 7S - 8$ bits below the top), smaller elements form further tiers. Every (band, tier, digit) pair is a plain INT8 GEMM whose int32 sums cannot overflow ($|P| \le K \cdot 96 \cdot 128 < 2^{26.4}$); the few terms of an output are added in a 128-bit integer and rounded once, with a certificate when a term falls below the 94-bit window (it never failed). BF16×BF16 is the same with both sides in digits. A correctly rounded result cannot depend on the summation order, the tiling, the batch or the machine.

On the real weights of layer 5 (experts 0–15, all three matrices, fetched by HTTP range requests; Student-t activations scaled by the layer's real norm weight), against what inference engines do (BF16 tensor cores, FP32 accumulation, split-K chosen by shape):

certified correctly rounded (48 matrices, 79.7 M outputs)	all
independent check (Python integers, raw bytes)	1500/1500
batch 1, 3, 17, 64, 200 vs 4096; decode kernel vs prefill	bit-identical
engine outputs not correctly rounded, split-K 1/2/4/8 (expert 0 w1)	81.5/67.1/55.6/51.2%
engine, split-K 1 vs 4: outputs whose bits differ	71%
prefill, 4096 tokens, 3072×3584: exact / BF16	8.0 / 2.9 ms
decode, 1 token: exact (packed 4-bit) / FP4 GEMV, FP32	0.062 / 0.125 ms

On the real router of the layer (896×7168 BF16, FP32 logits, sigmoid plus learned bias, top 16), the rounding of the logits alone sends ∼1 token in 13 000 to a different expert (11 in 147 456 token–split pairs), i.e. over K3's 92 MoE layers ∼1 token in 140 runs a different expert somewhere; with exact logits the choice depends on the input alone. A whole MoE block of layer 5 (62 real experts fetched on demand, the latent projections, RMSNorm and the shared experts; products correctly rounded, reductions in FP64 in a fixed order) gives the same bits for every token alone or in a batch, while the engine path changes 49% of the block's 28 672 BF16 outputs between split-K 1 and 4.

Where exactness costs nothing: a trillion-parameter model on this laptop is a streaming problem. Per text token K3 reads 25.8 GB of routed MXFP4 experts and 111.2 GB of dense BF16 weights (attention 72.4, shared experts 24.3; summed from the 96 safetensors headers); we measured 5.1 GB/s from the NVMe drive and 45 GB/s over PCIe 5 ×16, while the exact decode kernel consumes 94 GB/s. A bit-reproducible run therefore takes the same ∼27 s per token as an ordinary one, all of it reading. What this laptop lacks is 1.56 TB of storage (669 GB free).

The largest model with open training data. Nemotron 3 Ultra [20] publishes weights, recipe and pre-training data (560 524 578 816 parameters in the BF16 checkpoint's metadata; 14.8T tokens in 226 datasets). Its routed experts are NVFP4 for weights and activations: E2M1 values, one E4M3 scale per 16, an FP32 global scale. An NVFP4 block of 16 is exactly one `mma.m16n8k16` INT8 ($|p_b| \le 2304$), the product of the two E4M3 scales is an integer ≤ 225 times $2^{\varepsilon_W+\varepsilon_A}$, and after a shift of 18 every term is below 2^{47}: the $K/16 \le 320$ terms add in an `int64` without loss, and the 104-bit product with $g_W g_A$ is rounded once. On expert 0 of MoE layers 49 and 99 (real weights, 4096 tokens through the real latent projection, up, relu^2, down) the tensor-core path, an integer-loop path, 32 tokens alone, Python rationals from the raw bytes (sampled) and a CPU recomputation of all 29 360 128 outputs of layer 49 (BLAS block sums, Python integers) agree in every bit. The hardware's own NVFP4 instruction, `mma.m16n8k64.kind::mxf4nvf4.block_scale.scale_vec::4X` with FP32 accumulation (what inference engines run), leaves 18–49% of the outputs not correctly rounded and changes up to 0.81% of them between split-K 1 and 4; an emulation with an exact epilogue shows that the FP32 sum alone accounts for 0–0.9% and the rest is $\alpha = \mathrm{fp32}(g_W g_A)$. With the same plain tiling the native instruction runs at 50–55 TOPS and the exact path at 16–18 TOPS. The exact decode reads 405–487 GB/s of packed weights for one token and 136–184 GB/s for eight; the whole NVFP4 checkpoint (∼320 GB) fits this laptop's free disk.

10 Energy

Under these kernels the laptop holds ∼95 W and the SM clock drops to ∼1.8 GHz (2.76 GHz with only the FP64 units busy). For GEMM at $n = 4096$ (`bench/energy.cu`, power every 200 ms): native FP64 7.4 GFLOPS/W, Ozaki-II at 16 moduli 53.6 (7.2×), at 12 moduli 69.5. For a whole solve ($n = 4096$, $\kappa = 10^{20}$, 20 back-to-back GPU solves after a warm-up; CPU package power from the RAPL counters, GPU power from `nvidia-smi`):

	time	CPU package	GPU	energy per solve
`__float128` LU, 24 threads	55.0 s	106.5 W	8.4 W (idle)	6.3 kJ
dd LU on the GPU	0.238 s	28.2 W	68.3 W	**23 J**

That is ∼270× less energy per solve with both CPU and GPU counted (360× GPU alone against CPU package alone). DRAM power is not exposed on this laptop.

11 Limitations

The INT8 GEMMs are power-bound on this laptop, so a faster kernel does not translate into time; gains come from fewer operations (fewer moduli) and less data movement. The cost of the higher precisions grows roughly with the square of the number of slices, as MPFR's does at these sizes, so the ratio against MPFR does not grow without bound; it is largest against software emulation of each width. The triple-double solver is panel-bound at $n = 4096$. Kernel launches cost ∼7 μs under WSL2 (0.83 μs per node in a CUDA graph), which matters for the many small launches of the panel. All measurements are on one laptop model; the CPU references use the same machine's 24 threads.

12 Related work

Ozaki scheme II [5] emulates FP64 GEMM with INT8 and the CRT and reports 7.4–9.8 TFLOPS on an RTX 4090; RIKEN's GEMMul8 is its public code. Follow-up work covers SGEMM and power efficiency on an RTX 5080 [6], FP8 and FP4 variants [7, 8, 9], fused kernels reaching 81% of INT8 peak on Blackwell [10] and complex GEMM [11]. Multimodular exact integer matrix multiplication is classical (FLINT; conversions to and from a residue system written as matrix products by Doliskani, Giorgi, Lebreton and Schost [12]); Close in spirit: Ozaki-II for multiple-precision GEMM on CPU INT8 units (AMX, SVE2-i8mm) [13]; integer matrix products at various precisions on AMX and IFMA by van der Hoeven and Mezzarobba [14]; modular products over large prime fields on GPUs by Berthomieu, Graillat, Lesnoff and Mary [15]; EmuGEMM already fuses the modular reduction into the GEMM epilogue [10]. What we have not found is exact integer GEMM of 10^3–10^5-bit entries on GPU INT8 tensor cores, conversions included, measured against FLINT (to be confirmed by reading these works in full). what is new here is running it, conversions included, on INT8 tensor cores with byte-split residues. Ozaki-II mentions an extension beyond FP64 as possible; we found no implementation or measurement of a double-double or wider GEMM built this way, nor of linear solves at these condition numbers on this class of hardware. For low-precision neural networks, exact (Kulisch-style) accumulation was proposed in hardware designs [2, 3]; batch-invariant kernels [1] make LLM inference deterministic across batch sizes on one setup, and say explicitly that they are not hardware or software-version invariant. Correct rounding of every product, as in Section 9, is invariant to both, on commodity INT8 units. *(References to be re-checked at their source before submission.)*

13 Conclusion

On a laptop GPU whose FP64 units are 517 times slower than its INT8 tensor cores, precision becomes a software parameter: the same INT8 GEMMs yield FP64, double-double, triple-double and wider products, and an ill-conditioned dense solve that needs quad precision finishes 499× sooner than on the laptop's own CPU, with ∼270× less energy. Exact integer products of up to 131 063-bit entries come from the same units, 10–19× faster than FLINT. Exact determinants and exact rational solutions of int8 systems, where no big number has to enter the GPU, come 85–355× faster than FLINT, and with 32–256-bit entries high-order lifting on the same GEMMs keeps the ratio rising with the bits, up to 306×. The next steps are the panel (now the bound of

the triple-double solve), CUDA graphs for its launches, and out-of-core sizes beyond the GPU's 12 GB.

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